

Code Challenge

Toy Robot Simulator

Overview

The application is a simulation of a toy robot moving on a square tabletop, of dimensions 5 units x 5 units. There are no other obstructions on the table surface. The robot is free to roam around the surface of the table but must be prevented from falling to destruction. Any movement that would result in the robot falling from the table must be prevented, however further valid movement commands must still be allowed.

Requirements

Create a console application that can read in commands of the following form:

```
PLACE X,Y,F
```

```
MOVE
```

```
LEFT
```

```
RIGHT
```

```
REPORT
```

Command Descriptions

PLACE X,Y,F will put the toy robot on the table in position X,Y and facing NORTH, SOUTH, EAST or WEST. The origin (0,0) can be considered to be the SOUTH WEST most corner. It is required that the first command to the robot is a PLACE command, after that, any sequence of commands may be issued, in any order, including another PLACE command. The application should discard all commands in the sequence until a valid PLACE command has been executed.

MOVE will move the toy robot one unit forward in the direction it is currently facing.

LEFT and **RIGHT** will rotate the robot 90 degrees in the specified direction without changing the position of the robot.

REPORT will announce the X,Y and F of the robot. This can be in any form, but standard output is sufficient.

A robot that is not on the table can choose to ignore the MOVE, LEFT, RIGHT and REPORT commands.

Input can be from a file, or from standard input, as the developer chooses.

Constraints

The toy robot must not fall off the table during movement. This also includes the initial placement of the toy robot. Any move that would cause the robot to fall must be ignored.

Example Input and Output

Example 1

```
> PLACE 0,0,NORTH  
  
> MOVE  
  
> REPORT
```

Output:

```
0,1,NORTH
```

Example 2

```
> PLACE 0,0,NORTH  
  
> LEFT  
  
> REPORT
```

Output:

```
0,0,WEST
```

Example 3

```
> PLACE 1,2,EAST  
  
> MOVE  
  
> MOVE  
  
> LEFT  
  
> MOVE  
  
> REPORT
```

Output:

```
3,3,NORTH
```

Submission Guidelines

Please submit the following:

- Your complete source code, in a language of your choice.
- Test data to exercise the application.
- A short README with instructions on how to build and run your solution.
- A brief description of your development process. Tools used (What and how they were used), resources consulted, and any notable decisions or trade-offs you made along the way.
- If you have used AI to write any parts of the ToyRobot, please also submit any prompts or artefacts used as inputs to code generation.

Good luck! If you have any questions about the requirements, please don't hesitate to reach out.